

Redshift Spectrum: Demo

To Begin with the Lab

Summary of the Lab

In this lab, **Amazon Redshift Spectrum** was configured to query data stored directly in **Amazon S3** without loading it into **Amazon Redshift**. An **IAM** role with permissions for **Amazon S3** and **AWS Glue** was created and associated with the Redshift cluster. An external schema linked to the **AWS Glue Data Catalog** was created, followed by an external table that referenced a dataset stored in Amazon S3. The external metadata was automatically registered in the Glue Data Catalog, allowing Redshift to access the S3 data transparently. Finally, Spectrum queries were executed to retrieve records and join external S3 data with native Redshift tables, verifying successful integration.

Understanding Amazon Redshift Spectrum

Amazon Redshift Spectrum allows **Amazon Redshift** to query files stored in **Amazon S3** directly without importing the data into Redshift tables.

Use Case

Consider a company storing recent transactional data inside **Amazon Redshift** and historical sales data inside **Amazon S3**. Loading years of historical data into Redshift would increase storage costs and maintenance overhead.

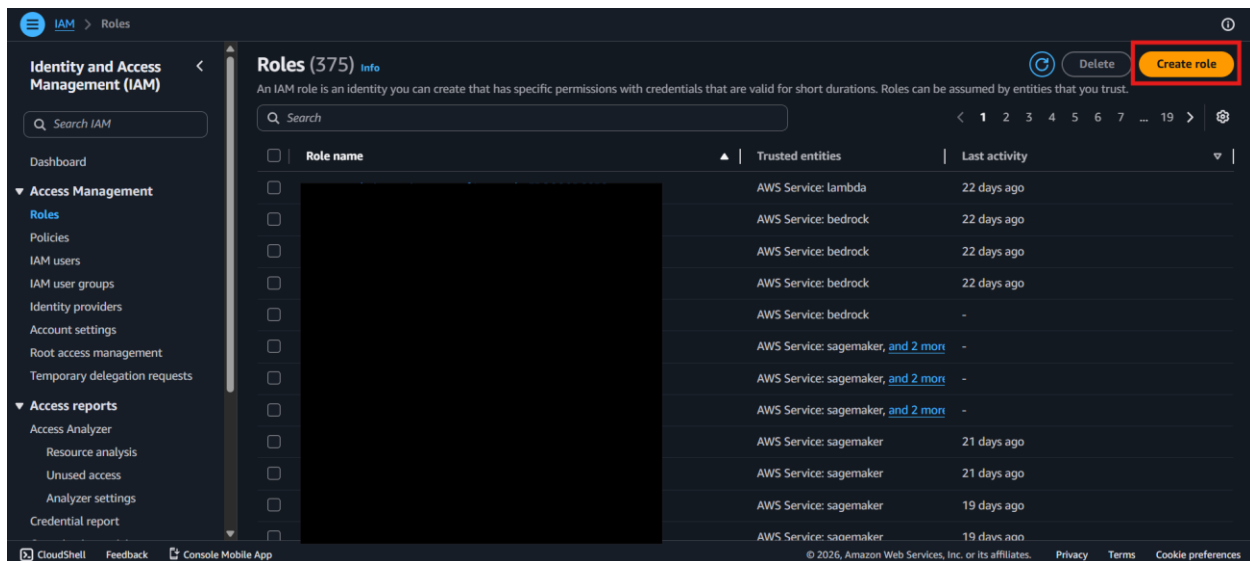
Using **Amazon Redshift Spectrum**, analysts can:

- Keep historical datasets in **Amazon S3**
- Query S3 files directly from Redshift
- Join S3 data with Redshift tables
- Reduce Redshift storage requirements
- Analyze large datasets without ETL loading processes

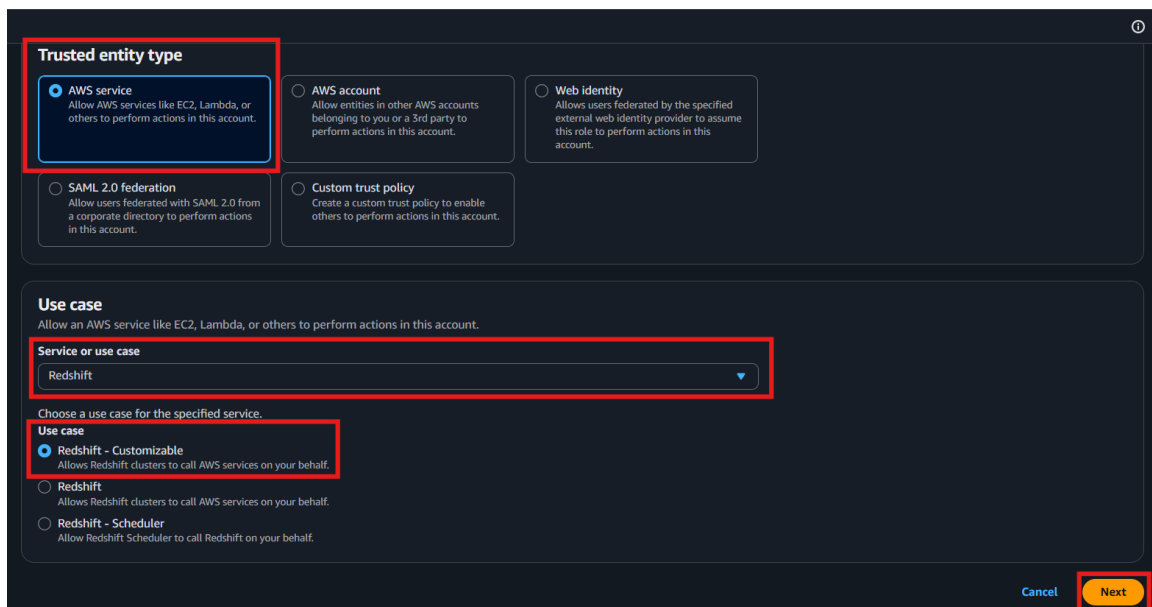
Lab Steps

- Sign in to the AWS Management Console

Create the IAM Role for Redshift Spectrum

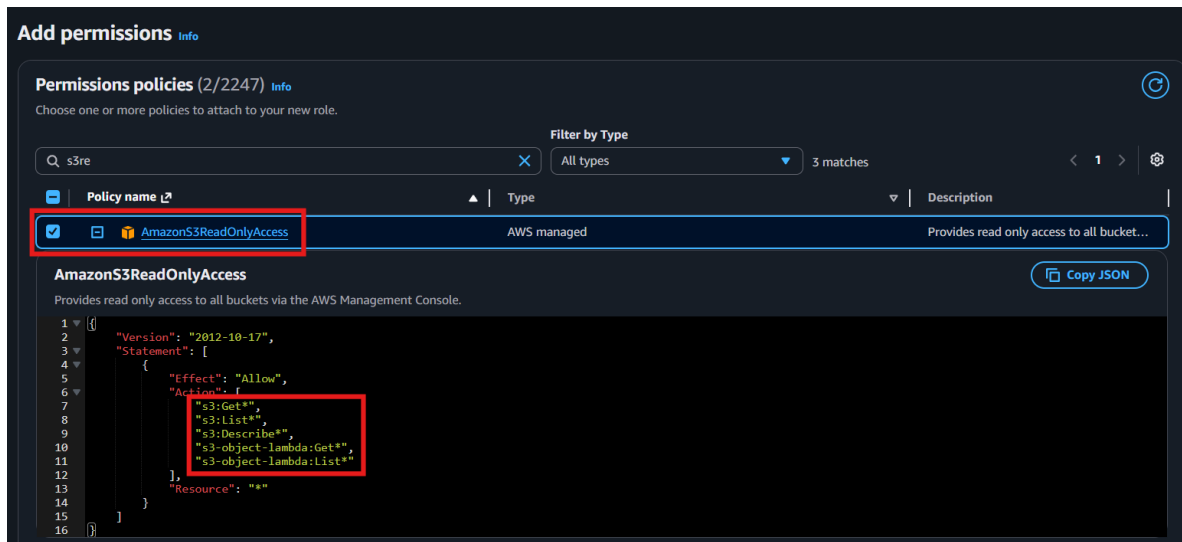


- Navigate to **IAM** → **Roles** → **Create role**
- Select **AWS service** as the trusted entity type
- Select **Amazon Redshift** as the service
- Choose **Customizable** use case
- Click **Next**

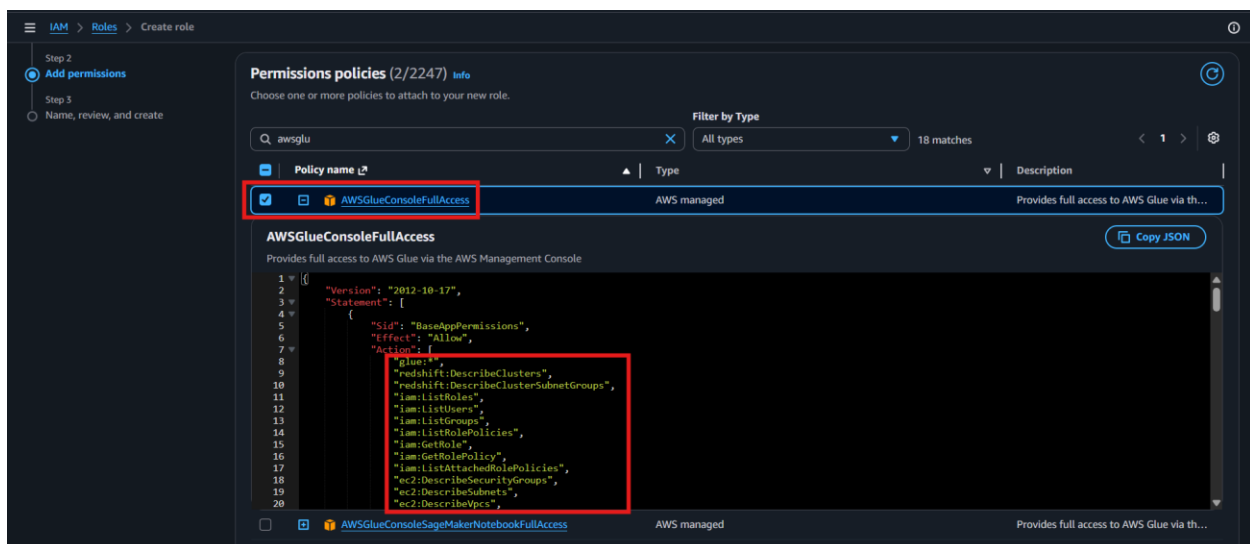


Add Required Permissions

- Search and select **AmazonS3ReadOnlyAccess**



- Search and select **AWSGlueConsoleFullAccess**



- Review the selected permissions
- Enter the role name as **redshift-spectrum-role**

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+,=,@,-_' characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-/\[\]!#\$%^&*()~''

Step 1: Select trusted entities

Trust policy

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": [  
7         "sts:AssumeRole"  
8       ],  
9       "Principal": {  
10        "Service": [  
11          "redshift.amazonaws.com"  
12        ]  
13      }  
14    ]  
15  }  
16 }
```

Click Create role

```
9   "Principal": {  
10    "Service": [  
11      "redshift.amazonaws.com"  
12    ]  
13  }  
14 }  
15 }  
16 }
```

Step 2: Add permissions

Permissions policy summary

Policy name ↗	Type	Attached as
AmazonS3ReadOnlyAccess	AWS managed	Permissions policy
AWSGlueConsoleFullAccess	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional [Info](#)
Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.
No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

Cancel

Previous

Create role

Save the Role ARN

- Open the newly created role
- Copy the role ARN for later use

Role redshift-spectrum-role created. [View role](#) ×

redshift-spectrum-role [Info](#) Delete

Allows Redshift clusters to call AWS services on your behalf.

Summary ARN copied Edit

Creation date
June 01, 2026, 14:28 (UTC+05:30)

Last activity
-

Maximum session duration
1 hour

[Permissions](#) [Trust relationships](#) [Tags](#) [Last Accessed](#) [Revoke sessions](#)

Permissions policies (2) [Info](#) Simulate Remove Add permissions

You can attach up to 10 managed policies.

Filter by Type All types < 1 > ⚙️

☐	Policy name	Type	Attached entities
☐	AmazonS3ReadOnlyAccess	AWS managed	25
☐	AWSGlueConsoleFullAccess	AWS managed	2

Associate the Role with the Redshift Cluster

- Navigate to **Amazon Redshift** → **Clusters**
- Select the target cluster
- Click **Actions** → **Manage IAM roles**

[Amazon Redshift](#) > [Clusters](#) > redshift-cluster-1

redshift-cluster-1 Actions Edit Add partner integration Query data

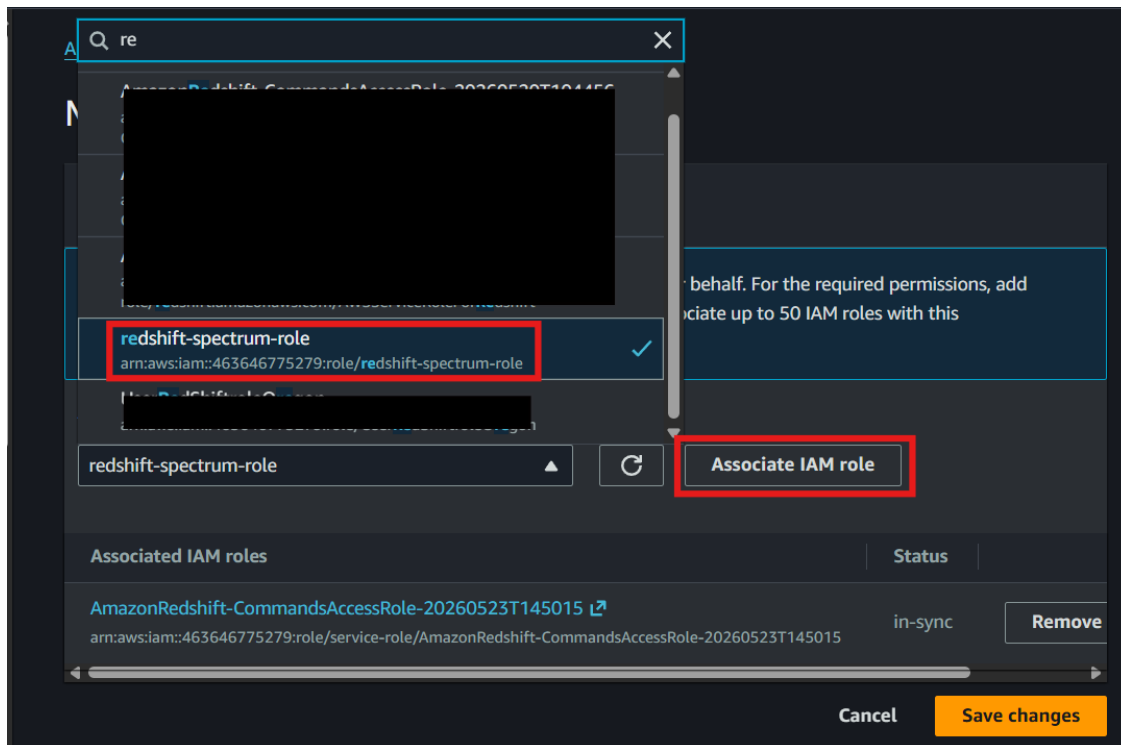
General information [Info](#)

Cluster identifier redshift-cluster-1	Status Available
Custom domain name -	Date created June 01, 2026, 14:31 IST
Cluster namespace ARN <code>arn:aws:redshift:ap-south-1:463646775279:namespace:4ebbcd42-015c-42d9-a353-33d3c3811228</code>	Multi-AZ No

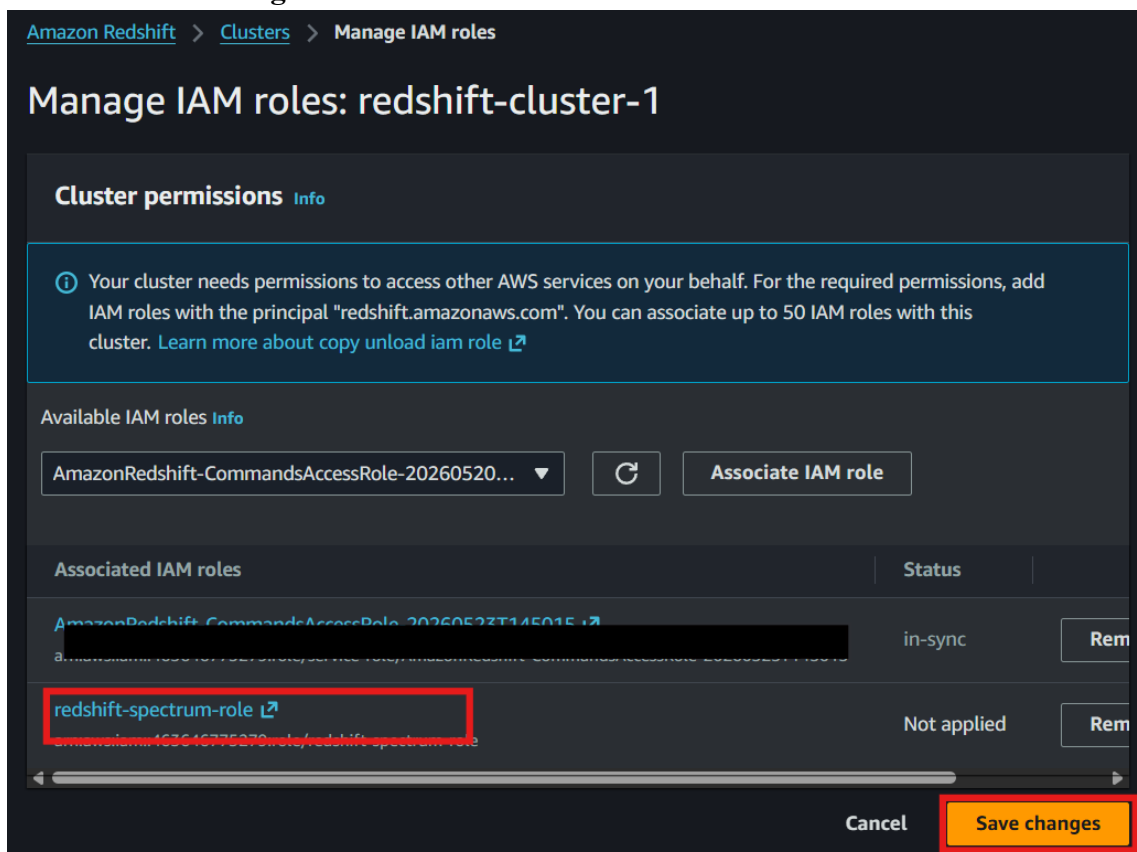
Actions Permissions Manage IAM roles

- Modify publicly accessible setting
- Register with AWS Glue Data Catalog
- Edit AWS Glue Data Catalog registration
- Deregister from AWS Glue Data Catalog
- Activate Multi-AZ
- Custom domain name
- Create custom domain name
- Backup and disaster recovery
- Restore table
- Create snapshot
- Configure cross-region snapshot
- Relocate
- Permissions
- Manage IAM roles
- Change admin user password
- Manage tags

- Select the newly created Spectrum role
- Click **Associate IAM role**



- Click **Save changes**



- Wait for the cluster update to complete

Open Query Editor v2

- Navigate to **Amazon Redshift** → **Query data**
- **Open Query Editor v2**

The screenshot shows the Amazon Redshift console interface for a cluster named 'redshift-cluster-1'. The breadcrumb navigation at the top reads 'Amazon Redshift > Clusters > redshift-cluster-1'. On the right side, there are buttons for 'Actions', 'Edit', 'Add partner integration', and 'Query data' (highlighted in orange). Below these buttons is a dropdown menu with options: 'Query in query editor', 'Query in query editor v2' (highlighted with a red box), and 'Query in Amazon SageMaker Unified Studio'. The main content area is titled 'General information' and contains a table with the following details:

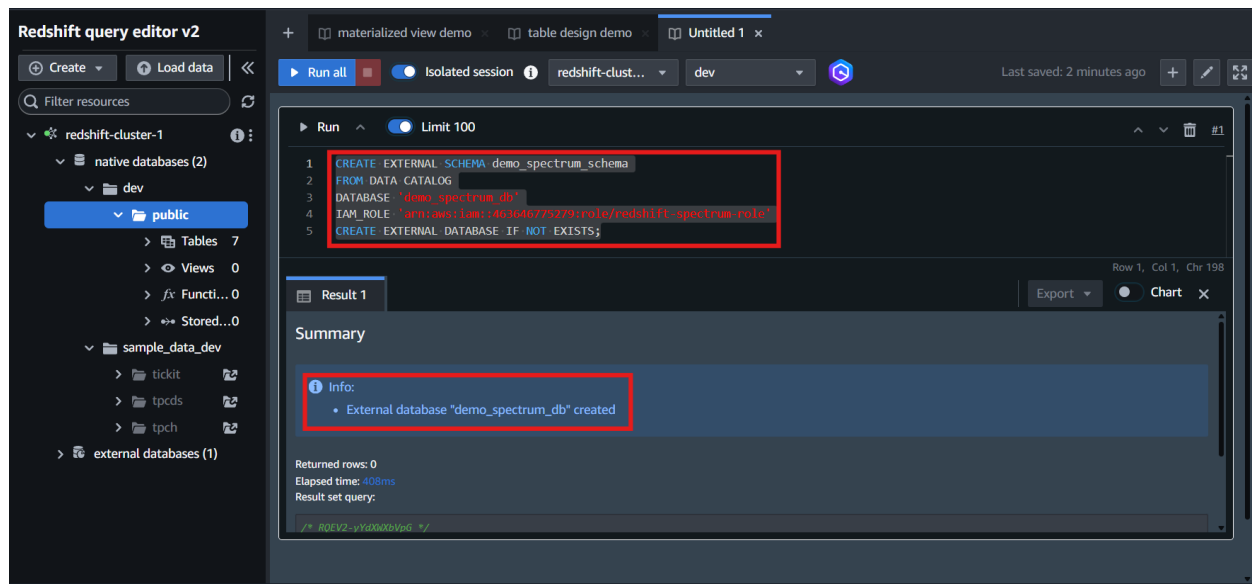
Cluster identifier redshift-cluster-1	Status Available	Node type ra3.large	Endpoint redshift-cluster-1.cowmxpc1gjl.ap-south-1.redshift.amazonaws.com:5439/dev
Custom domain name -	Date created June 01, 2026, 14:31 IST	Number of nodes 1	JDBC URL jdbc:redshift://redshift-cluster-1.cowmxpc1gjl.ap-south-1.redshift.amazonaws.com:5439/dev Download driver
Cluster namespace ARN arn:aws:redshift:ap-south-1:463646775279:namespace:4ebbcd42-015c-42d9-a353-33d3c3811228	Multi-AZ No	Patch version Patch 202	ODBC URL Driver={Amazon Redshift (x64)}; Server=redshift-cluster-1.cowmxpc1gjl.ap-south-1.redshift.amazonaws.com; Database=dev

- Connect to the target database

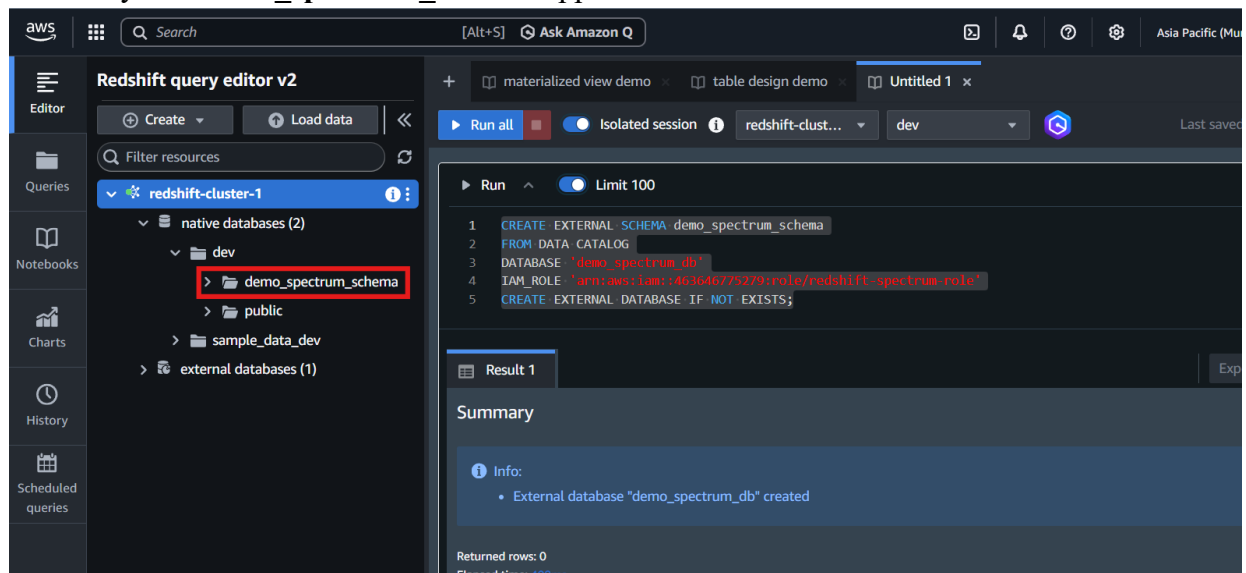
Create the External Schema

- Execute the following query to create an external schema linked to the **AWS Glue Data Catalog**

```
CREATE EXTERNAL SCHEMA demo_spectrum_schema
FROM DATA CATALOG
DATABASE 'demo_spectrum_db'
IAM_ROLE 'arn:aws:iam::<account-id>:role/redshift-spectrum-role'
CREATE EXTERNAL DATABASE IF NOT EXISTS;
```



- Run the query
- Refresh the database explorer
- Verify that **demo_spectrum_schema** appears in the database



Upload Sample Data to Amazon S3

- Navigate to **Amazon S3**
- Open the target bucket
- Upload the sales dataset file
- Verify the file is available in the desired folder location

Create the External Table

- Execute the external table creation query

```

CREATE EXTERNAL TABLE demo_spectrum_schema.sales (
  salesid integer,
  listid integer,

```


sellerid integer,
buyerid integer,
eventid integer,
dateid smallint,
qtysold smallint,
pricepaid decimal(8,2),
commission decimal(8,2),
saletime timestamp

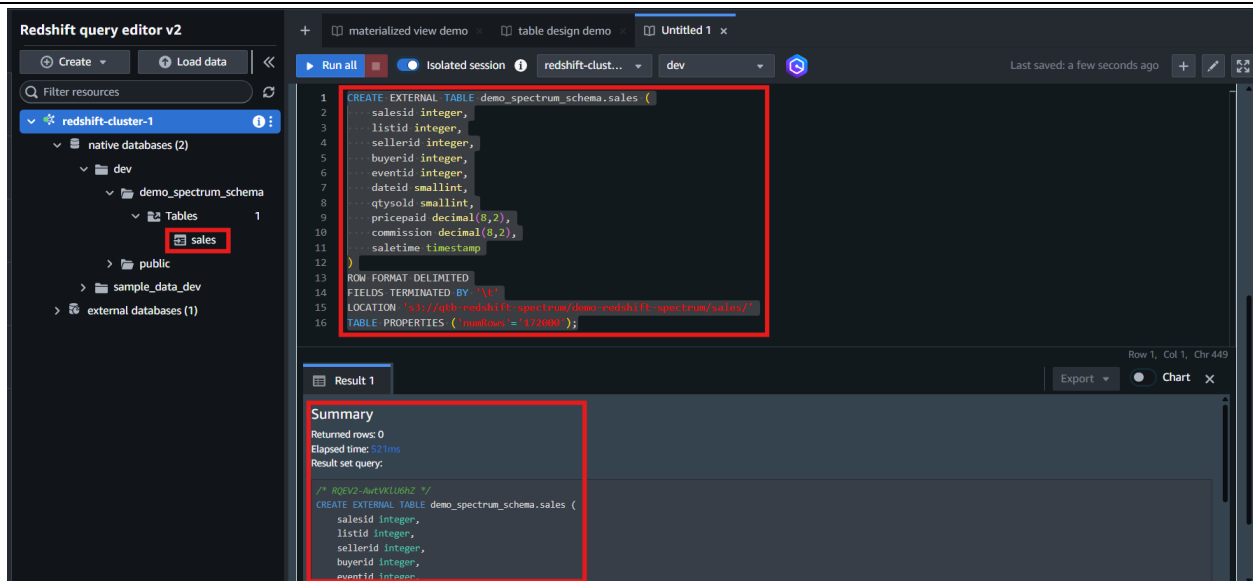
)

ROW FORMAT DELIMITED

FIELDS TERMINATED BY '\t'

LOCATION 's3://YourBucketName/sales/'

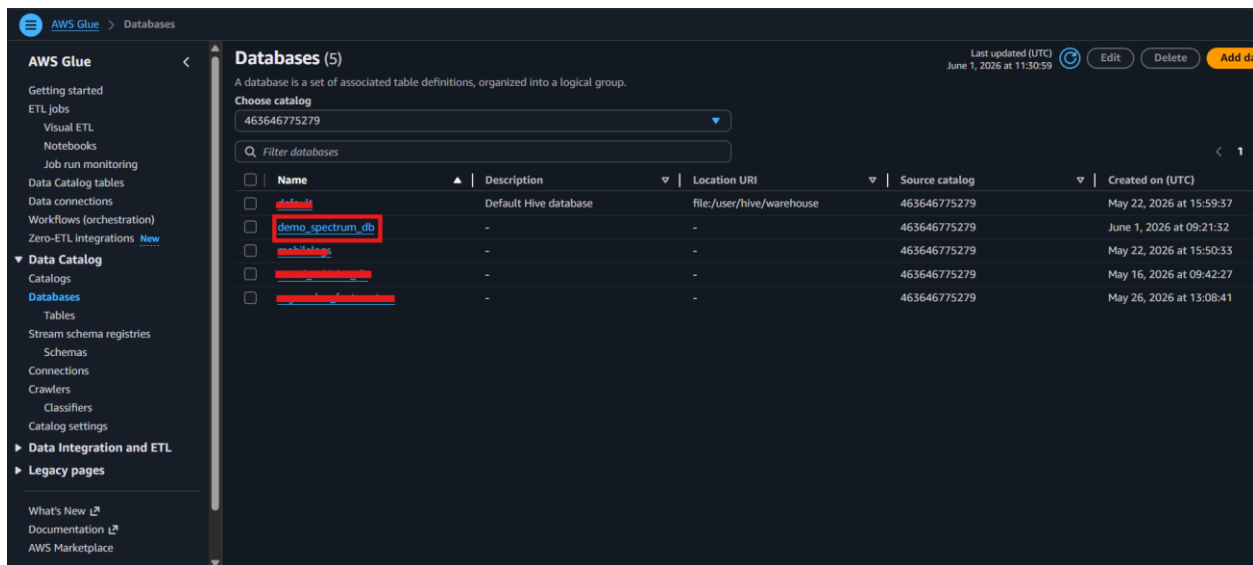
TABLE PROPERTIES ('numRows'='172000');



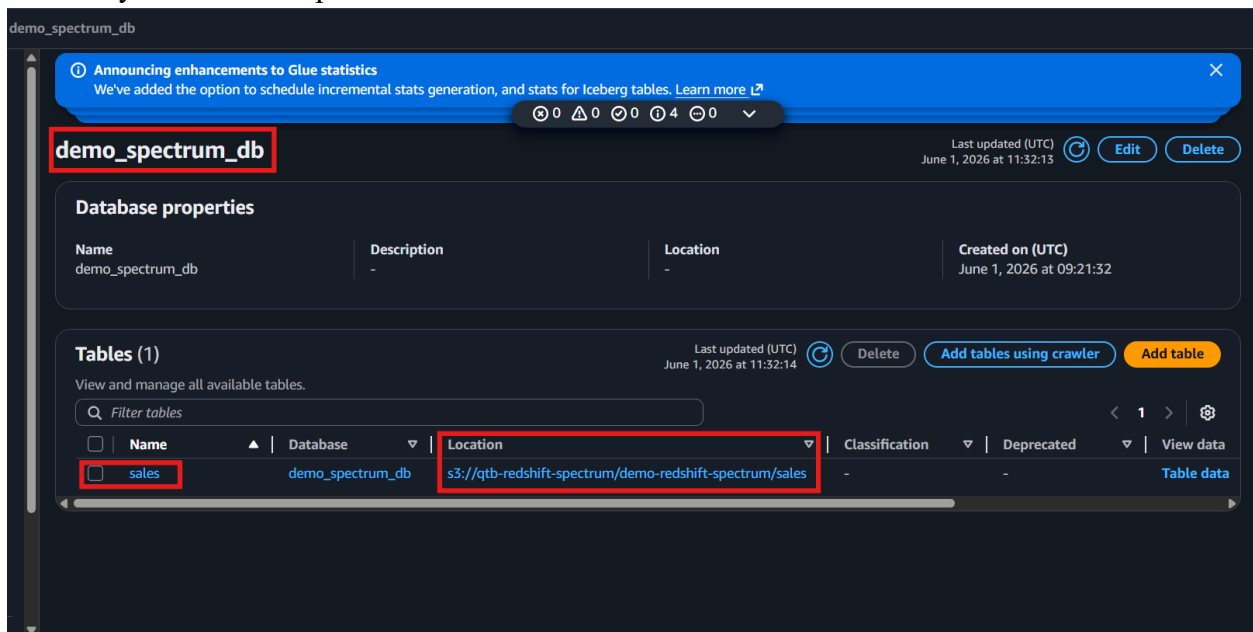
- Run the query
- Refresh the schema explorer
- Verify that the **sales** table is created under **demo_spectrum_schema**

Verify AWS Glue Catalog Registration

- Navigate to **AWS Glue** → **Data Catalog** → **Databases**



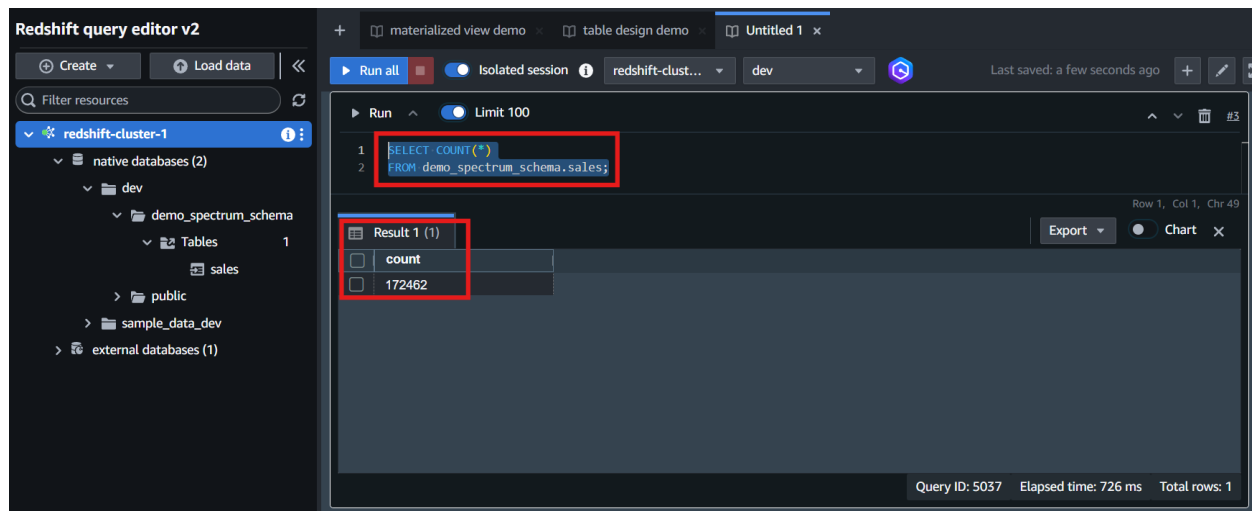
- Verify that **demo_spectrum_db** is created automatically
- Open the database
- Verify that the **sales** table exists
- Verify that the table points to the correct **Amazon S3** location



Query Data Stored in Amazon S3

- Execute the record count query

```
SELECT COUNT(*)
FROM demo_spectrum_schema.sales;
```

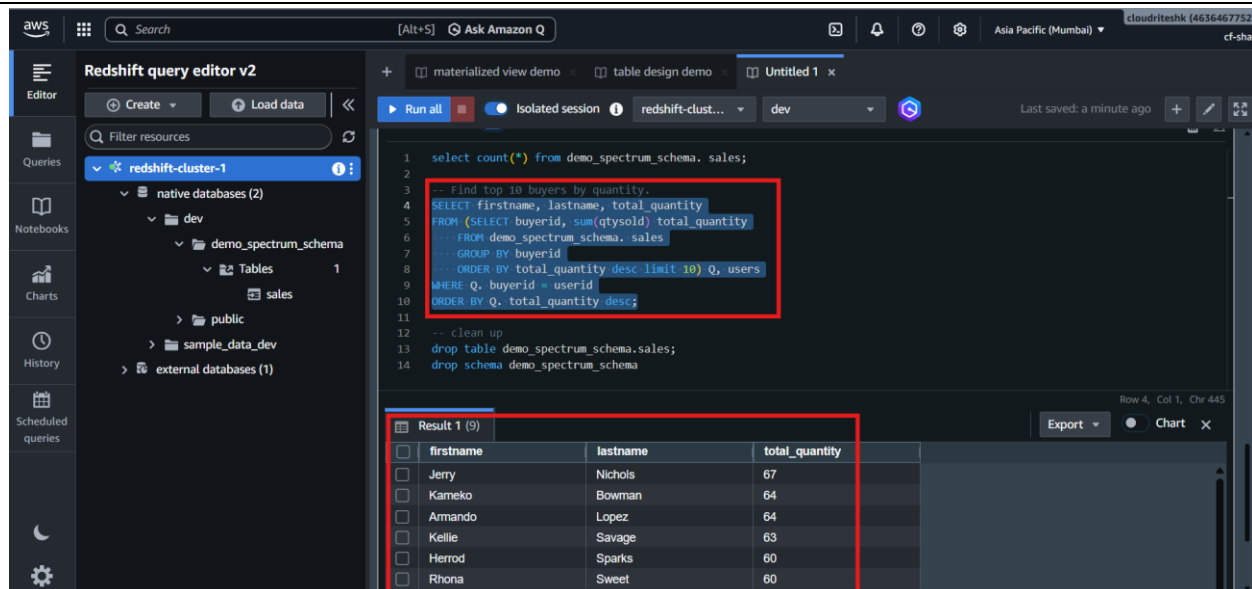


- Run the query
- Verify that records are returned successfully

Query External and Internal Tables Together

- Execute a join query between the external sales table and a Redshift users table

```
SELECT firstname, lastname, total_quantity
FROM (SELECT buyerid, sum(qtysold) total_quantity
      FROM demo_spectrum_schema. sales
      GROUP BY buyerid
      ORDER BY total_quantity desc limit 10) Q, users
WHERE Q. buyerid = userid
ORDER BY Q. total_quantity desc;
```



- Run the query
- Verify that results are returned successfully

Verify Spectrum Functionality

- Confirm that data was queried directly from **Amazon S3**
- Confirm that the external table behaves like a native Redshift table
- Confirm that joins between external and internal tables execute successfully
- Verify that **Amazon Redshift Spectrum** is functioning correctly